PRODUCT USER MANUAL

For OMI_EXTREME_STATE_GLOBAL_trend

Issue: 2.0

Contributors : Cécile Béchonnet (CLS)

Approval Date: November 2025

MERCATOR OCEAN
INTERNATIONAL
2 avenue de l'Aérodrome de Montaudran, 31400 Toulouse, FRANCE
Tél : +33 5 61 39 38 02 - Fax : +33 5 61 39 38 99
Société civile de droit français au capital de 2 000 000 € - 522 911 577 RCS Toulouse - SIRET 522 911 577 00024

marine.copernicus.eu
mercator-ocean.eu

RECORD TABLE

Issue	Date	§	Description of Change	Author	Validated By
1.0	2024/04/22	all	Creation of the document	C. Béchonnet	P.Zunino
2.0	2025/04/22	all	Add time series uncertainties.Add a new dataset with global trend maps. Rename product and datasets	C.Béchonnet	P.Zunino

TABLE OF CONTENTS

RECORD TABLE **2**

GLOSSARY AND ABBREVIATIONS **4**

DATA ACCESS **5**

1) INTRODUCTION **6**

a) Summary 6

2) DESCRIPTION OF THE PRODUCT SPECIFICATION **7**

a) General Information 7

b) Production System Description 8

c) Dataset details 8

3) FILE FORMAT **10**

4) FILES NOMENCLATURE **10**

5) REFERENCES **10**

GLOSSARY AND ABBREVIATIONS

CMEMS	Copernicus Marine Environment Monitoring Service
CLS	Collecte Localisation Satellite
MOI	Mercator Ocean International
NetCDF	Network Common Data Form
OMI	Ocean Monitoring Indicator
PUM	Product User Manual
QUID	Quality Information Document
SWH	Significant Wave Height
L4	Level-4
JFM	January-February-March
JAS	July-August-September

DATA ACCESS

After registration, you will be able to download our data. To assist you, our HelpCenter is available, and more specifically its section about download.

Information on operational issues on products and services can be found on our User Notification Service. If needed, you can contact us .

1) INTRODUCTION

a) Summary

This document is the user manual for the CMEMS Ocean Monitoring Indicator (OMI) - OMI_EXTREME_STATE_trend.
This OMI provides two datasets : omi_extreme_state_global_trend_mean_and_extreme which gives global maps of trends of significant wave height's (SWH) seasonal mean and extreme values. The second dataset is omi_extreme_state_global_trend_area_averaged_mean_and_extreme which provides time series in three oceanic regions of SWH seasonal mean and extreme as well as their trends from 2002 to 2020. These estimates are derived from the reprocessed global L4 SWH product (WAVE_GLO_PHY_SWH_L4_MY_014_007, see this OMI QUID). The extreme SWH is defined as the 95th percentile of the daily maximum of SWH over the chosen period and region. The 95th percentile represents the value below which 95% of the data points fall, indicating higher wave heights than usual. The mean and the 95th percentile of SWH are calculated for two periods of the year to take into account the seasonal variability of waves (January, February, and March, and July, August, and September) and are in metres while the trends are in cm yr^{-1}.

The product is available online and disseminated through the CMEMS Information System. Files downloaded are in NetCDF format and follow CF-1.7 convention.

This OMI is described in the Quality Information Document (QUID) CMEMS-WAV-OMI-QUID-EXTREME_STATE.pdf (see section 5).

Information on operational issues on products and services can be found on our **User Notification Service**. If you have any questions, please **contact us**.

2) DESCRIPTION OF THE PRODUCT SPECIFICATION

a) General Information

Product(s) name	OMI_EXTREME_STATE_GLOBAL_trend
Geographical coverage	Global with 3 regional boxes: - North Atlantic: 30°N-43°N; 52°W-18°W - South of Australia: 60°S-35°S; 110°E-155°EAntarctic Circumpolar Current: 66°S-47°S; 38°W-42°E
Variables	vavh_seasonal_mean vavh_seasonal_mean_uncertainty_min vavh_seasonal_mean_uncertainty_max vavh_seasonnal_q95 vavh_seasonal_q95_uncertainty_min vavh_seasonal_q95_uncertainty_max vavh_trend_mean vavh_trend_q95 vavh_trend_mean_uncertainty vavh_trend_q95_uncertainty vavh_seasonal_nbr vavh_trend_seasonal_mean_global vavh_trend_seasonal_q95_global vavh_trend_seasonal_uncertainty_mean_global vavh_trend_seasonal_uncertainty_q95_global
Available time series	From 2002 to 2020
Temporal resolution	Seasonal value
Horizontal resolution	2° x 2°
Integration depth (for example)	surface
Format	NetCDF CF1.7

b) Production System Description

Seasonal (JFM and JAS) SWH mean and 95th percentile (maps and time series for each of the considered regions) were obtained from the daily SWH L4 MY product (WAVE_GLO_PHY_SWH_L4_014_007) over the period 2002-2020. Linear regression was used to recover the trends for both variables.
In the time series, the trend and uncertainties were obtained from the linear regression, while the uncertainty given with the mean and 95th percentile values of the time series was constructed using bootstrap. The error is given at the 95% confidence level.

For the maps, the p-value obtained from the linear regression was used to obtain the statistical significance of the trends obtained. If the p-value is less than 0.05, the trend is considered significant.

c) Dataset details

omi_extreme_state_global_trend_area_averaged_mean_and_extreme contains the time series of seasonal values of the SWH mean and 95th percentile.
vavh_seasonal_mean [m] Seasonal average of the SWH defined over the chosen region sea_surface_wave_significant_height_seasonal_mean **vavh_seasonal_q95 [m]** Seasonal 95th percentile of the SWH defined over the chosen region. sea_surface_wave_significant_height_seasonal_95percentile **vavh_seasonal_mean_uncertainty_min [m]** Minimal bound of the uncertainty on the seasonal average of SWH over the chosen region. sea_surface_significant_wave_height_mean_uncertainty_minimal_bound **vavh_seasonal_mean_uncertainty_max [m]** Maximal bound of the uncertainty on the seasonal average of SWH over the chosen region. sea_surface_significant_wave_height_mean_uncertainty_maximal_bound **vavh_seasonal_q95_uncertainty_min [m]** Minimal bound of the uncertainty on the seasonal 95th percentile of SWH over the chosen region. sea_surface_significant_wave_height_95percentile_uncertainty_minimal_bound **vavh_seasonal_q95_uncertainty_max [m]** Maximal bound of the uncertainty on the seasonal 95th percentile of SWH over the chosen region.

sea_surface_significant_wave_height_95percentile_uncertainty_maximal_bound **vavh_seasonal_nbr [-]** Average number of observations considered each season. sea_surface_significant_height_seasonal_number_of_observations **vavh_trend_mean [cm yr-1]** Trend of average SWH over the chosen region sea_surface_wave_significant_height_mean_trend **vavh_trend_q95 [cm yr-1]** Trend of 95th percentile SWH over the chosen region. sea_surface_wave_significant_height_95percentile_trend **vavh_trend_mean_uncertainty [cm yr-1]** Uncertainty on the trend of average SWH. sea_surface_significant_height_trend_uncertainty_mean **vavh_trend_q95_uncertainty [cm yr-1]** Uncertainty on the trend of the 95th percentile of SWH. sea_surface_significant_height _trend_uncertainty_95percentile

omi_extreme_state_global_trend_mean_and_extreme contains the global trends of seasonal values of the SWH mean and 95th percentile
vavh_trend_seasonal_mean_global [cm yr-1] Trend of global seasonal average of SWH. sea_surface_significant_wave_height_seasonal_mean_trend **vavh_trend_seasonal_q95_global [cm yr-1]** Trend of global seasonal 95th percentile of SWH. sea_surface_significant_wave_height_seasonal_95percentile_trend **vavh_trend_seasonal_uncertainty_mean_global [-]** P-Value of global trends of seasonal average of SWH (significant if <0.05). sea_surface_significant_wave_height_seasonal_mean_trend_uncertainty **vavh_trend_seasonal_uncertainty_q95_global [-]** P-Value of global trends of seasonal 95th percentile of SWH (significant if <0.05) sea_surface_significant_wave_height_seasonal_95percentile_trend_uncertainty

Note that the standard name and units of each of these variables in **OMI_EXTREME_STATE_GLOBAL_trend** product are not yet included in CF conventions since they were created for this contribution.

3) FILE FORMAT

The products are stored using the NetCDF format.

To know more about the NetCDF format, please follow this link:
What is the format of Copernicus Marine products ? NetCDF

To understand the differences between netCDF and Zarr, please consult this article:
how-to-choose-between-netcdf-and-zarr-format-using-the-toolbox

4) FILES NOMENCLATURE

Information about nomenclature of files when downloaded can be found in this article: " How is defined the nomenclature of Copernicus Marine data? | Copernicus Marine Help Center "

OMI_EXTREME_STATE_GLOBAL_trend files nomenclature when downloaded through the CMEMS Marine Data Store Services is based on the following nomenclature:
<datasetname>_<timestep>_P<productiontime>.nc
Where :

- <datasetname> is the dataset reference (OMI_EXTREME_STATE_GLOBAL_ trend_mean_and_extreme or OMI_EXTREME_STATE_GLOBAL_area_averaged_mean_and_extreme).
- <timestep> is the first timestep contained in the file.
- <productiontime> is the production time when the file was generated.
- .nc is the standard NetCDF filename extension.

Example:
OMI_EXTREME_STATE_GLOBAL_trend_mean_and_extreme_2002_P20250502.nc

5) REFERENCES

Quality Information Document

See the Quality Information Document (QUID) of products CMEMS-WAV-OMI-QUID-EXTREME_STATE-V2.0 and WAVE_GLO_PHY_SWH_L4_MY_014_007